Breast Cancer Genes and Clinical Data

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Datasets Introduction

- Two tables available: gene expression (GeneX) and clinical data (clinical_data).
- Same 1,231 patients represented in both tables.
- Target outcome: `vital_status`, a binary status (Alive / Dead).

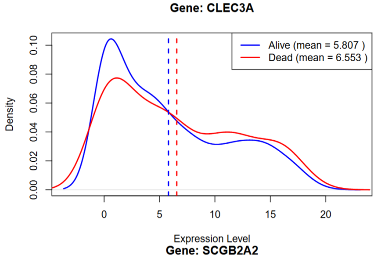
Genes Data

- GeneX: 1,231 rows (patients) and 5,000 columns (genes).
- Rows: patient IDs (e.g., EW-A2FS-01A-11R-A17B).
- Columns: gene names (e.g., CLEC3A), top 5,000 most variable genes.
- Cells: continuous expression values per patient-gene pair.

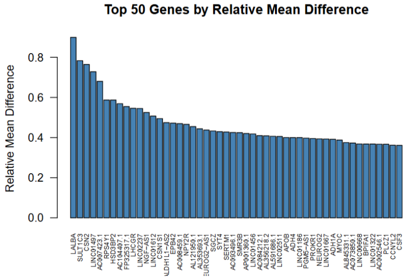
```
##                         CLEC3A   SCGB2A2      CPB1    GSTM1      TFF1
## EW-A2FS-01A-11R-A17B  0.000000 16.259357 18.018521  0.00000 13.656872
## OL-A6VR-01A-32R-A33J  0.000000 16.522045  7.189825 12.77992 12.802920
## E9-A226-01A-21R-A157 10.915132  5.285402  5.930737 10.22279 10.195987
## A8-A08H-01A-21R-A00Z  3.169925  9.952741  7.734710  0.00000  1.584963
## D8-A27H-01A-11R-A16F  5.000000  7.643856  7.467606  1.00000  0.000000
## D8-A3Z6-01A-11R-A239 15.954469 13.971723 13.489346  1.00000 16.881711
```

Genes Exploration

- Plotted, for each gene, the density of (Dead vs. Alive) and averaged these curves.
- Computed relative differences of mean expression between Alive and Dead groups.



- Ranking genes by relative mean difference highlighted the most discriminative markers.



Clinical Data Exploration

- clinical_data: 1,231 rows and 24 columns.
- Rows: patients (aligned with GeneX).
- Columns: clinical variables (e.g., initial_weight, sites_of_involvement, vital_status).
- Cells: clinical values per patient (integer, float, character or NA).

	initial_weight	tissue_type	laterality
	<numeric>	<character>	<character>
EW-A2FS-01A-11R-A17B	110	Tumor	Left
OL-A6VR-01A-32R-A33J	380	Tumor	Right
E9-A226-01A-21R-A157	510	Tumor	Left
A8-A08H-01A-21R-A00Z	250	Tumor	Left
D8-A27H-01A-11R-A16F	170	Tumor	Right
D8-A3Z6-01A-11R-A239	60	Tumor	Left

Modeling Plan

- **Genes (GeneX):** LASSO logistic regression (L1) to handle p $\gg$ n and perform feature selection/dimensionality reduction.

$$\text{LASSO:}\ \min_{\beta} \sum_{i=1}^{n} \log\left(1 + e^{-y_i x_i^T \beta}\right) + \lambda \|\beta\|_1$$

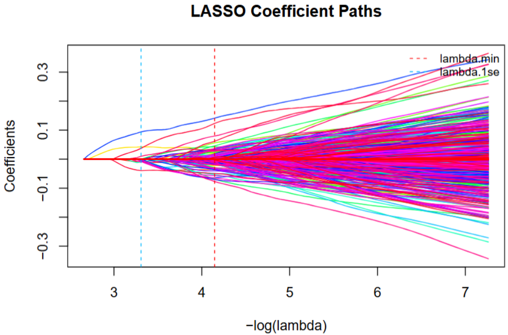
- **Clinical data:** RIDGE logistic regression (L2) to shrink coefficients without enforcing sparsity.

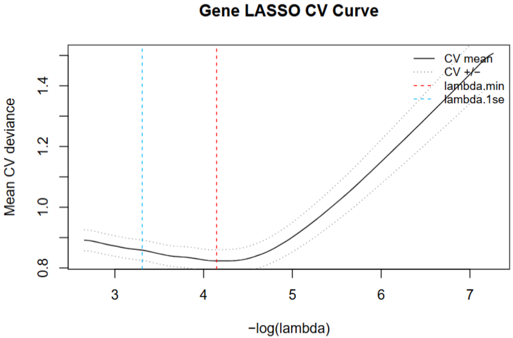
$$\text{RIDGE:}\ \min_{\beta} \sum_{i=1}^{n} \log\left(1 + e^{-y_i x_i^T \beta}\right) + \lambda \|\beta\|_2^2$$

- **Both tables together:** Elastic Net (L1+L2) to keep some selection while stabilizing correlated predictors.

$$\text{Elastic Net:}\ \min_{\beta} \sum_{i=1}^{n} \log\left(1 + e^{-y_i x_i^T \beta}\right) + \lambda\left(\alpha \|\beta\|_1 + \tfrac{1-\alpha}{2} \|\beta\|_2^2\right)$$

LASSO Genes

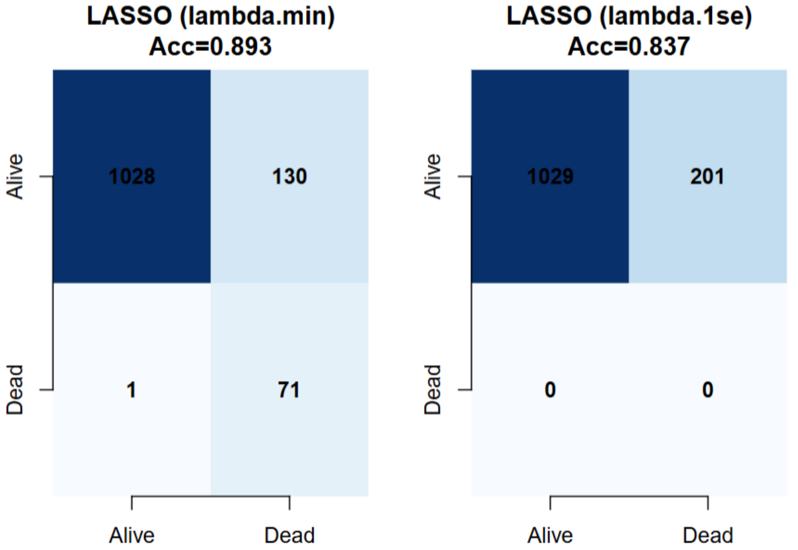




Most important genes variables based on their λ value using $\lambda_{\min}$.

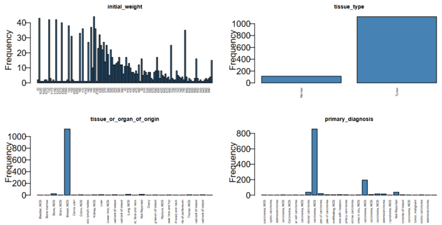
```
##  LINC01235       PSD2     SLC6A3     SFTPA1 AC017048.3        TNN     GPS2P1
## 0.15446630 0.13872679 0.09444156 0.09151916 0.08350079 0.08297162 0.07585189
## AL031432.2     MPPED1     RPS4Y1     OR56A3     GSTT2B     SEMA3B
## 0.07167134 0.06407371 0.06336668 0.06124477 0.06046761 0.05848547
```

LASSO Genes


LASSO (lambda.min)
Acc=0.893
Alive
Dead
1028
130
1
71
Alive
Dead
LASSO (lambda.1se)
Acc=0.837
Alive
Dead
1029
201
0
0
Alive
Dead

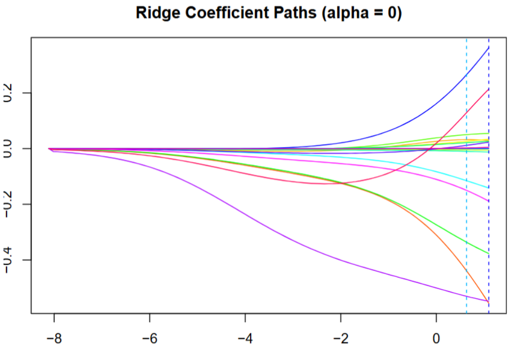
RIDGE Clinical Data

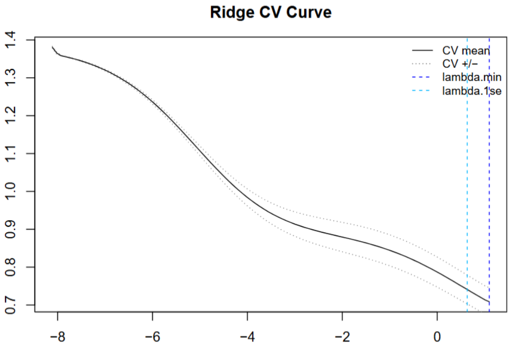
- We first performed data cleaning and inspection.
- After analysis we removed redundant variables: `age_at_diagnosis` and `days_to_birth` (both reflect age information captured by `age_at_index`), and `bcr_patient_barcode` (acts as the row index).
- We also removed `disease_type` and `primary_site` since they contained a single identical observation for all patients.
- We encoded categorical variables but kept NA values as explicit missing values for modeling.




```
race => 1:american indian or alaska native, 2:asian, 3:black or african american, 4:not reported
gender => 1:female, 2:male, 3:NA
ethnicity => 1:hispanic or latino, 2:not hispanic or latino, 3:not reported, 4:Unknown, 5:NA
vital_status => 1:Alive, 2:Dead, 3:NA
age_is_obfuscated => 1:FALSE, 2:TRUE, 3:NA
```

RIDGE Clinical Data

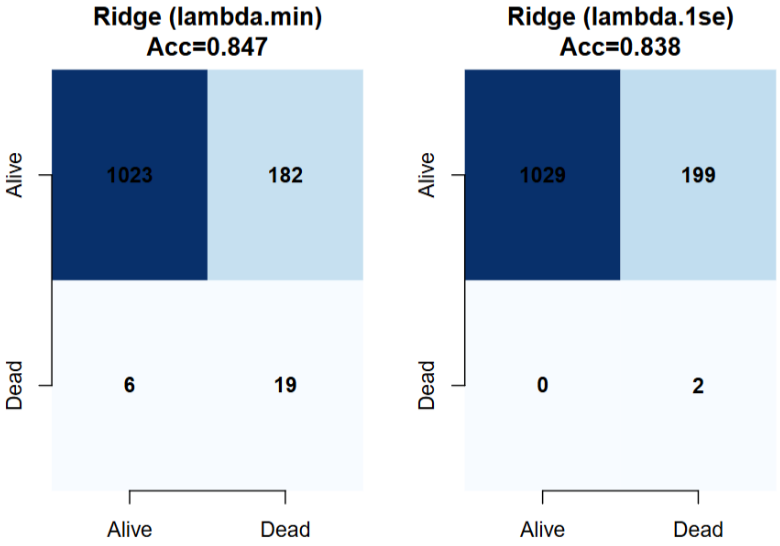




Most important clinical variables based on their λ value using λ_{min}.

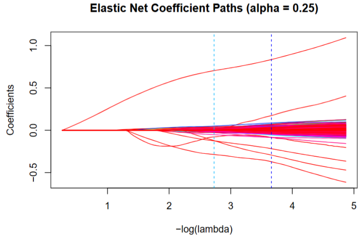
```
##                  tissue_type                        gender
##                    0.5557430                     0.5481611
## diagnosis_is_primary_disease  follow_ups_disease_response
##                    0.3768788                     0.3621960
##              age_is_obfuscated
##                    0.2135292
```

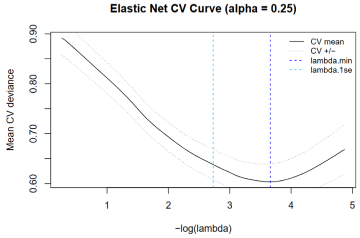
RIDGE Clinical Data



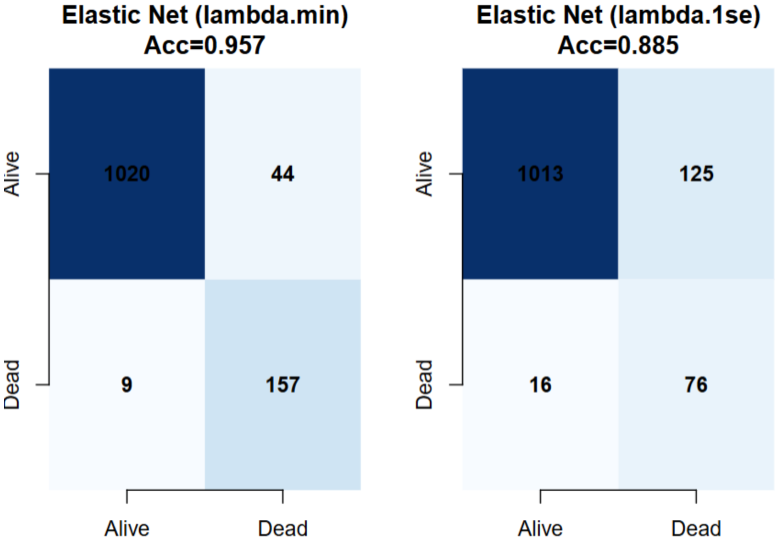
Elastic Net Genes + Clinical Data

- We combined genes and clinical datasets by taking the intersection on patient IDs to ensure aligned rows.
- Elastic Net mixes L1 and L2 penalties; we selected the mixing parameter α by cross-validation choosing among: 0, 0.25, 0.5, 0.75 and 1.
- This lets us balance sparsity (LASSO) and stability (RIDGE) when predictors are correlated.





Elastic Net Genes + Clinical Data



Conclusion

- Using Elastic Net (tuning α by CV) allowed balancing sparsity and stability across predictors.
- We tested $\alpha = 0,\ 0.25,\ 0.5,\ 0.75,\ \text{and } 1.$
- The best performing α was 0.25, indicating clinical predictors contributed more in the combined model.
- Combined data improved predictive performance compared to using genes or clinical data alone.